

Problem Description:

Tina owns a match making company, which even to her surprise is an extreme hit. She says that her success rate cannot be matched (Yes, letterplay!) in the entire match-making industry. She follows an extremely simple algorithm to determine if two people are matches for each other. Her algorithm is not at all complex, and makes no sense - not even to her. But she uses it anyway.

Let's say that on a given day she decides to select n people - that is, n boys and n girls. She gets the list of n boys and n girls in a random order initially. Then, she arranges the list of girls in ascending order on the basis of their height and boys in descending order of their heights. A girl A_i can be matched to a boy on the same index only, that is, B_i and no one else. Likewise, a girl standing on A_k can be only matched to a boy on the same index B_k and no one else.

Now to determine if the pair would make an ideal pair, she checks if the modulo of their heights is 0, i.e., $A_i \% B_i == 0$ or $B_i \% A_i == 0$. Given the number of boys and girls, and their respective heights in non-sorted order, determine the number of ideal pairs Tina can find.

Constraints:

1 <= Test Cases <= 10^2

1 <= N <= 10^4

1 <= A_i, B_i <= 10^5

Input Format:

The first line contains number of test cases. Then, the next line contains an integer, n , saying the number of boys and girls. The next line contains the height of girls, followed by the height of boys.

Output Format:

Print the number of ideal pairs in a separate lines

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

2
4
1 6 9 12
4 12 3 9
4
2 2 2 2
2 2 2 2

EXPECTED OUTPUT

2
4

Test Case 2

INPUT (STDIN)

2
4
3 6 9 12
4 12 16 9
4
1 1 1 1
2 2 2 2

EXPECTED OUTPUT

3
4

▼ Mandatory Test Cases

Test Case 1

KEYWORD

for(int i = 0;i<n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

210

Test Case 3

NLOC

23

```
Sorting1.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int cmp_asc(const void *a, const void *b) {
5     return (*(int*)a - *(int*)b);
6 }
7
8 int cmp_desc(const void *a, const void *b) {
9     return (*(int*)b - *(int*)a);
10}
11
12int main() {
13    int t;
14    if (scanf("%d", &t) != 1) return 0;
15    while (t--) {
16        int n;
17        scanf("%d", &n);
18        int a[10005], b[10005];
19        for (int i = 0; i < n; i++) {
20            scanf("%d", &a[i]);
21        }
22        for (int i = 0; i < n; i++) {
23            scanf("%d", &b[i]);
24        }
25        qsort(a, n, sizeof(int), cmp_asc);
```

```
26     qsort(b, n, sizeof(int), cmp_desc);
27     int count = 0;
28     for (int i = 0; i < n; i++) {
29         if (a[i] % b[i] == 0 || b[i] % a[i] == 0) {
30             count++;
31         }
```

```
32     }  
33     printf("%d\n", count);  
34 }  
35 return 0;  
36}  
37
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

4

1 6 9 12

4 12 3 9

4

2 2 2 2

2 2 2 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting1.exe

2

4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2

4

3 6 9 12

4 12 16 9

4

1 1 1 1

2 2 2 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting1.exe

3

4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

Question description

Sajid is an third year student in a reputed institution.

Although he scored well in many subjects, he did not an expert in computer programming languages.

But Sajid's computer examination is scheduled for next week.

As per the blueprint, many questions would come from the sorting topic.

He collected previous year's questions. one of the repeated questions is to sort the given set of numbers using Selection Sort

The first line of the input contains the number of elements N , the second line of the input contains the numbers A_i to be sorted.

In the output print the the final sorted array in the given format.

Can you help him ?

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq A_i \leq 10^9$$

Input:

The first line of the input contains the number of elements

the second line of the input contains the numbers to be sorted.

Output:

print the the final sorted array in the given format.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
15 17 11 25 1

EXPECTED OUTPUT

1 11 15 17 25

Test Case 2

INPUT (STDIN)

6
65 45 27 21 25 11

EXPECTED OUTPUT

11 21 25 27 45 65

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void selectionSort(int arr[],int n)

Test Case 2

KEYWORD

void swap(int *xp,int *yp)

Test Case 3

KEYWORD

void printArray(int arr[],int size)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

47

```
1 #include <stdio.h>
2
3 void swap(int *xp, int *yp) {
4     int temp = *xp;
5     *xp = *yp;
6     *yp = temp;
7 }
8
9 void selectionSort(int arr[], int n) {
10     int i, j, min_idx;
11     for (i = 0; i < n - 1; i++) {
12         min_idx = i;
13         for (j = i + 1; j < n; j++) {
14             if (arr[j] < arr[min_idx]) {
15                 min_idx = j;
16             }
17         }
18         swap(&arr[min_idx], &arr[i]);
19     }
20 }
21
```

main | CH.SC.U4CSE25225 | Sorting2.c

Ln 41, Col 1 Spaces: 4 UTF-8 CRLF C

```
22 void printArray(int arr[], int size) {
23     for (int i = 0; i < size; i++) {
24         printf("%d", arr[i]);
25         if (i < size - 1) printf(" ");
26     }
27 }
```

```
26     }
27     printf("\n");
28 }
29
30 int main() {
31     int n;
32     if (scanf("%d", &n) != 1) return 0;
33     int arr[100005];
```

```
34     for (int i = 0; i < n; i++) {
35         scanf("%d", &arr[i]);
36     }
37     selectionSort(arr, n);
38     printArray(arr, n);
39     return 0;
40 }
41
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
5
```

```
15 17 11 25 1
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting2.exe
```

```
1 11 15 17 25
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
6
```

```
65 45 27 21 25 11
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting2.exe
```

```
11 21 25 27 45 65
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Sorting	Question Information	Level 1	Challenge 13
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Problem Description:

Siva has several containers, each with a number of fruits in it. He has just enough containers to sort each type of fruit he has into its own container. Siva wants to sort the fruits using his sort method.

Siva wants to perform some number of swap operations such that:

- Each container contains only fruits of the same type.
- No two fruits of the same type are located in different containers.

Function Description

organizingContainers has the following parameter(s):

int container[n][m]: a two dimensional array of integers that represent the number of balls of each color in each container

- Constraints:
- $1 \leq q \leq 10$
 - $1 \leq n \leq 100$
 - $0 \leq \text{containers}[i][j] \leq 10^9$

Input Format:

The first line contains an integer 'q', the number of queries.

- Each of the next 'q' sets of lines is as follows:
- The first line contains an integer 'n', the number of containers (rows) and ball types (columns).
- Each of the next 'n' lines contains 'n' space-separated integers describing row containers[i].

Output Format:

For each query, print Possible on a new line if David can satisfy the conditions above for the given matrix. Otherwise, print Impossible.

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
2
1 1
1 1
2
0 2
1 1

EXPECTED OUTPUT

Possible
Impossible

Test Case 2

INPUT (STDIN)

2
2
2 1
1 2
2
1 2
2 1

EXPECTED OUTPUT

Possible
Possible

Mandatory Test Cases

Test Case 1

KEYWORD

void insertionSort(long int *p, long int n)

Test Case 2

KEYWORD

for(i=0; i<n; i++)

Test Case 3

KEYWORD

while(q--)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

5

Test Case 2

TOKEN COUNT

450

Test Case 3

NLOC

63

Sorting3.c +

```
1 #include <stdio.h>
2
3 void insertionSort(long int *p, long int n) {
4     long int i, j, key;
5     for (i = 1; i < n; i++) {
6         key = p[i];
7         j = i - 1;
8         while (j >= 0 && p[j] > key) {
9             p[j + 1] = p[j];
10            j = j - 1;
11        }
12        p[j + 1] = key;
13    }
14}
15
16int main() {
17    int q;
18    if (scanf("%d", &q) != 1) return 0;
19    while (q--) {
20        long int n, i, j;
21        scanf("%ld", &n);
22        long int row_sum[105] = {0};
23        long int col_sum[105] = {0};
24        for (i = 0; i < n; i++) {
25            for (j = 0; j < n; j++) {
```

```
26         long long val;
27         scanf("%lld", &val);
28         row_sum[i] += val;
29         col_sum[j] += val;
30     }
31 }
32 insertionSort(row_sum, n);
33 insertionSort(col_sum, n);
34 int possible = 1;
35 for (i = 0; i < n; i++) {
36     if (row_sum[i] != col_sum[i]) {
37         possible = 0;
```

```
38         break;
39     }
40 }
41 if (possible) {
42     printf("Possible\n");
43 } else {
44     printf("Impossible\n");
45 }
46 }
47 return 0;
48}
49
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2
```

```
2
```

```
1 1
```

```
1 1
```

```
2
```

```
0 2
```

```
1 1
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting3.exe
```

```
Possible
```

```
Impossible
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
2
```

```
2 1
```

```
1 2
```

```
2
```

```
1 2
```

```
2 1
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting3.exe
```

```
Possible
```

```
Possible
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Sorting	Question Information	Level 1	Challenge 14
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Question description

APPU needed a laptop, so he went to a neighbouring garage sale.

At a sale, there were n laptops.

The cost of a laptop with index i is ai rupees.

Some laptops have a negative price, meaning their owners are willing to pay APPU if he buys their broken laptop. APPU is free to purchase whichever laptop he desires.

Despite his strength, he can only handle m laptops at a time, and he has no desire to return to the auction.

Please, help APPU find out the maximum sum of money that he can earn.

Constraints:

$1 \leq T \leq 10$

$1 \leq n, m \leq 100$

$-1000 \leq a_i \leq 1000$

Input Format:

First line of the input contains T denoting the number of test cases.Each test case has 2 lines :

first line has two spaced integers n m.

second line has n integers [a0...ai...an-1].

Output Format:

The maximum sum of money that LALU can earn, given that he can carry at most m Laptops.

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
5 3
-6 0 35 -2 4

EXPECTED OUTPUT

8

Test Case 2

INPUT (STDIN)

2
6 2
-4 5 0 -1 2 1
7 4
6 1 0 -2 2 -4 3

EXPECTED OUTPUT

5
6

Mandatory Test Cases

Test Case 1

KEYWORD

int MEGA_SALE(int arr[],int
no,int k)

Test Case 2

KEYWORD

void bubble_sort(int
arr[],int no)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

55

Sorting4.c +

```
1 #include <stdio.h>
2
3 void bubble_sort(int arr[], int no) {
4     int i, j, temp;
5     for (i = 0; i < no - 1; i++) {
6         for (j = 0; j < no - i - 1; j++) {
7             if (arr[j] > arr[j + 1]) {
8                 temp = arr[j];
9                 arr[j] = arr[j + 1];
10                arr[j + 1] = temp;
11            }
12        }
13    }
14}
15
16int MEGA_SALE(int arr[], int no, int k) {
17    bubble_sort(arr, no);
18    int earned = 0;
19    for (int i = 0; i < no && i < k; i++) {
20        if (arr[i] < 0) {
21            earned += (-arr[i]);
22        }
23    }
24    return earned;
25}
```

main | CH.SC.U4CSE25225 | Sorting4.c

Ln 41, Col 1 Spaces: 4 UTF-8 CRLF C

```
26
27int main() {
28    int t;
29    if (scanf("%d", &t) != 1) return 0;
30    while (t--) {
31        int n, m;
32        scanf("%d %d", &n, &m);
33        int arr[105];
```

```
34     for (int i = 0; i < n; i++) {
35         scanf("%d", &arr[i]);
36     }
37     printf("%d\n", MEGA_SALE(arr, n, m));
38 }
39 return 0;
40}
41
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

1
5 3
-6 0 35 -2 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting4.exe

8

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2
6 2
-4 5 0 -1 2 1
7 4
6 1 0 -2 2 -4 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting4.exe

5

6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Sorting	Question Information	Level 1	Challenge 15
Problem Description: One day Priya was jogging and realized that her life was boring. Everything was red, even the streets in the best park were red. Therefore she decided to make streets a little bit brighter. She know that every street in the park is a segment laying on the X axis with coordinates X_l, X_r ($X_l \leq X_r$). Streets may intersect or overlap. She chooses any subset of streets and paints them in green. After that she wants to get one continuous green segment. As she really likes number L the length of this segment has to be equal to L. Your task is to determine if it is possible to choose some subset of streets and paint them to get one green segment with the length equal to L? Constraints: $1 \leq \text{sum of all } N \leq 2 * 10^3$ $1 \leq L \leq 10^6$ $1 \leq X_l \leq X_r \leq 10^6$ $1 \leq N \leq 20, 1 \leq X_l \leq X_r \leq 200$, holds for test cases worth 10% of the problem's score. $1 \leq N \leq 100, 1 \leq X_l \leq X_r \leq 200$, holds for test cases worth 20% of the problem's score. Input Format: The first line contains one integer T - the number of test cases. Each test case starts with two integers N and L, denoting the number of streets and Priya's favorite number L. The next N lines contain two integers X_l, X_r, denoting the left and right borders of the street. Output Format: Print the output in a separate lines "Yes" if it is possible to paint some street and "No" otherwise.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

2
5 3
1 2
2 3
3 4
1 5
2 6
2 3
1 2
2 6

EXPECTED OUTPUT

Yes
No

Test Case 2

INPUT (STDIN)

2
5 3
1 4
2 3
3 5
5 1
5 6
2 3
2 4
1 3

EXPECTED OUTPUT

Yes
Yes

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t--)
```

Test Case 2

KEYWORD

```
for(ll i=0;i<n;i++)
```

Test Case 3

KEYWORD

```
for(ll j=0;j<n;j++)
```

Test Case 4

KEYWORD

```
if(cur_right==maxright)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

11

Test Case 2

TOKEN COUNT

289

Test Case 3

NLOC

48

```
Sorting5.c +
1 #include <stdio.h>
2
3 typedef long long ll;
4
5 struct Street {
6     ll l;
7     ll r;
8 };
9
10 int main() {
11     int t;
12     if (scanf("%d", &t) != 1) return 0;
13     while (t--) {
14         ll n, L;
15         scanf("%lld %lld", &n, &L);
16         struct Street streets[2005];
17         for (ll i = 0; i < n; i++) {
18             scanf("%lld %lld", &streets[i].l, &streets[i].r);
19             if (streets[i].l > streets[i].r) {
20                 ll tmp = streets[i].l;
21                 streets[i].l = streets[i].r;
22                 streets[i].r = tmp;
23             }
24         }
25         int possible = 0;
```

```
26     for (ll i = 0; i < n; i++) {
27         ll start = streets[i].l;
28         ll target = start + L;
29         ll cur_right = start;
30         while (cur_right < target) {
31             ll maxright = cur_right;
32             for (ll j = 0; j < n; j++) {
33                 if (streets[j].l <= cur_right && streets[j].r <= target) {
34                     if (streets[j].r > maxright) {
35                         maxright = streets[j].r;
36                     }

```

main | CH.SC.U4CSE25225 | Sorting5.c

Ln 97, Col 1 Spaces: 4 UTF-8 CRLF C

```
37         }
38     }
39     if (cur_right == maxright) {
40         break;
41     }
42     cur_right = maxright;
43 }
44 if (cur_right == target) {
45     possible = 1;
46     break;
47 }
```

```
48     }
49     if (possible) {
50         printf("Yes\n");
51     } else {
52         printf("No\n");
53     }
54 }
55 return 0;
56}
57
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

5 3

1 2

2 3

3 4

1 5

2 6

2 3

1 2

2 6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting5.exe

Yes

No

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2

5 3

1 4

2 3

3 5

5 1

5 6

2 3

2 4

1 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting5.exe

Yes

Yes

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```


DS	Session	Sorting	Question Information	Level 1	Challenge 16
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Question description

Selvan studies, engineering as per his father's wishes, while Aaron, whose family is poor, studies engineering to improve his family's financial situation. sumanth, however, studies engineering of his simple passion for developing data structure applications.

sumanth is participating in a hackathon for data structure application development.

sumanth task is to use Insertion Sort to sort the supplied set of numbers.

As a result, The input provides the number of components on the first line and the numbers to be sorted on the second line. Print the array's state at the third iteration and the final sorted array in the supplied format in the output.

Judge will determine whether the outcome is correct or not.

Can you help him ?

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq A_i \leq 10^9$$

Input Format:

The first line of the input contains the number of elements

the second line of the input contains the numbers to be sorted.

Output Format:

Fist line indicates print the status of the array at the 3rd iteration

second line print the the final sorted array in the given format.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
64 25 22 90 35

EXPECTED OUTPUT

22 25 64 90 35
22 25 35 64 90

Test Case 2

INPUT (STDIN)

5
16 21 22 10 35

EXPECTED OUTPUT

16 21 22 10 35
10 16 21 22 35

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void insertionSort(int
arr[],int n)

Test Case 2

KEYWORD

void printArray(int arr[],int
n)

Test Case 3

KEYWORD

insertionSort(arr, n);

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

50

Sorting6.c +

```
1 #include <stdio.h>
2
3 void printArray(int arr[], int n) {
4     for (int i = 0; i < n; i++) {
5         printf("%d", arr[i]);
6         if (i < n - 1) printf(" ");
7     }
8     printf("\n");
9 }
10
11 void insertionSort(int arr[], int n) {
12     int i, key, j;
13     for (i = 1; i < n; i++) {
14         key = arr[i];
15         j = i - 1;
16         while (j >= 0 && arr[j] > key) {
17             arr[j + 1] = arr[j];
18             j = j - 1;
19         }
```

main | CH.SC.U4CSE25225 | Sorting6.c

Ln 38, Col 1 Spaces: 4 UTF-8 CRLF C

```
21     if (i == 2) {
22         printArray(arr, n);
23     }
24 }
25 }
```

```
26
27int main() {
28    int n;
29    if (scanf("%d", &n) != 1) return 0;
30    int arr[100005];
31    for (int i = 0; i < n; i++) {
32        scanf("%d", &arr[i]);
```

```
33     }  
34     insertionSort(arr, n);  
35     printArray(arr, n);  
36     return 0;  
37 }  
38
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
5
64 25 22 90 35
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting6.exe
22 25 64 90 35
22 25 35 64 90
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
5
16 21 22 10 35
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting6.exe
16 21 22 10 35
10 16 21 22 35
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Sorting	Question Information	Level 1	Challenge 17
Question Description: Simon is studying B.Tech.-Mechanical Engineering. He's going to attend a computer science-based subject exam this semester. Due to the less preparation in the previous monthly tests, his internal mark decreased. His computer science Professor made an offer one more chance to boost up his internal marks. Professor assigns a program to Simon for the internal mark bootup. So Simon wants to solve the given task is, Given two arrays, A and B, of equal size n, the task is to find the minimum value of $A[0] * B[0] + A[1] * B[1] + \dots + A[n-1] * B[n-1]$, where shuffling of elements of arrays A and B is allowed. can you help him in solving Questions ? Constraints: $1 \leq T \leq 100$ $1 \leq N \leq 50$ $1 \leq A[i] \leq 20$ Input Format:					

The first line of input contains an integer denoting the no of test cases.

Then T test cases follow. Each test case contains three lines.

The first line of input contains an integer N denoting The size of the arrays.

In the second line are N space separated values of the array A[], and in the last line are N space separated values of the array B[].

Output Format:

For each test case in a new line print the required result.

Example :

Input : A[] = {3, 1, 1} and B[] = {6, 5, 4}.

Output : 23 Minimum value of $S = 1*6 + 1*5 + 3*4 = 23$.

Input : A[] = { 6, 1, 9, 5, 4 } and B[] = { 3, 4, 8, 2, 4 }

Output : 80. Minimum value of $S = 1*8 + 4*4 + 5*4 + 6*3 + 9*2 = 80$.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
4
2 5 3 7
8 5 7 3
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

Test Case 2

INPUT (STDIN)

```
2
3
13 11 1
6 15 14
5
6 11 9 15 4
3 41 8 21 4
```

EXPECTED OUTPUT

```
247
451
```

EXPECTED OUTPUT

83
247
451

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
void sort(int a[],int n,int  
flag)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

1

Test Case 2

TOKEN COUNT

420

Test Case 3

NLOC

55

```
Sorting7.c +
1 #include <stdio.h>
2
3 void sort(int a[], int n, int flag) {
4     for (int i = 0; i < n - 1; i++) {
5         for (int j = 0; j < n - i - 1; j++) {
6             if (flag == 1) {
7                 if (a[j] > a[j + 1]) {
8                     int tmp = a[j];
9                     a[j] = a[j + 1];
10                    a[j + 1] = tmp;
11                }
12            } else {
13                if (a[j] < a[j + 1]) {
14                    int tmp = a[j];
15                    a[j] = a[j + 1];
16                    a[j + 1] = tmp;
17                }
18            }
19        }
20    }
21}
22
23int main() {
```

main | CH.SC.U4CSE25225 | Sorting7.c

Ln 46, Col 1 Spaces: 4 UTF-8 CRLF C

```
25     if (scanf("%d", &t) != 1) return 0;
```

```
26  while (t--) {
27      int n;
28      scanf("%d", &n);
29      int a[55], b[55];
30      for (int i = 0; i < n; i++) {
31          scanf("%d", &a[i]);
32      }
33      for (int i = 0; i < n; i++) {
34          scanf("%d", &b[i]);
35      }
36      sort(a, n, 1);
```

```
37     sort(b, n, 0);
38     long long sum = 0;
39     for (int i = 0; i < n; i++) {
40         sum += (long long)a[i] * b[i];
41     }
42     printf("%lld\n", sum);
43 }
44 return 0;
45}
46
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
3
```

```
4
```

```
2 5 3 7
```

```
8 5 7 3
```

```
3
```

```
13 11 1
```

```
6 15 14
```

```
5
```

```
6 11 9 15 4
```

```
3 41 8 21 4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting7.exe
```

```
83
```

```
247
```

```
451
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
3
```

```
13 11 1
```

```
6 15 14
```

```
5
```

```
6 11 9 15 4
```

```
3 41 8 21 4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting7.exe
```

```
247
```

```
451
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Sorting	Question Information	Level 1	Challenge 18
----	---------	---------	----------------------	---------	--------------

Question description

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

2
5 6
2 5 4 5 2
5 3
1 6 3 5 2

EXPECTED OUTPUT

-1
3

Test Case 2

INPUT (STDIN)

2
5 6
12 15 14 15 12
5 3
11 61 31 51 21

EXPECTED OUTPUT

9
58

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void sort(int a[],int n)

Test Case 2

KEYWORD

for(i=0;i<n-1;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

5

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

40

Sorting8.c +

```
1 #include <stdio.h>
2
3 void sort(int a[], int n) {
4     int i, j, temp;
5     for (i = 0; i < n - 1; i++) {
6         for (j = 0; j < n - i - 1; j++) {
7             if (a[j] > a[j + 1]) {
8                 temp = a[j];
9                 a[j] = a[j + 1];
10                a[j + 1] = temp;
11            }
12        }
13    }
14}
15
16int main() {
17    int t;
18    if (scanf("%d", &t) != 1) return 0;
```

main | CH.SC.U4CSE25225 | Sorting8.c

Ln 36, Col 1 Spaces: 4 UTF-8 CRLF C

```
20     int n, k;
21     scanf("%d %d", &n, &k);
22     int a[1005];
23     for (int i = 0; i < n; i++) {
24         scanf("%d", &a[i]);
25     }
```

```
26     sort(a, n);
27     int max_dist = a[n - 1];
28     if (max_dist <= k) {
29         printf("-1\n");
30     } else {
31         printf("%d\n", max_dist - k);
```

```
32     }  
33 }  
34 return 0;  
35}  
36
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

5 6

2 5 4 5 2

5 3

1 6 3 5 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting8.exe

-1

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

2

5 6

12 15 14 15 12

5 3

11 61 31 51 21

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting8.exe

9

58

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Sorting	Question Information	Level 1	Challenge 19
----	---------	---------	----------------------	---------	--------------

Problem Description:

Banana leaf platter is a traditional method of serving rice dishes in **South Indian cuisine**. Due to the migration of South Indians, banana leaf rice can also be found in areas with significant ethnic South Indian diaspora such as **Malaysia** and **Singapore**.

Irfan is a banana leaf sales person.
he has N stacks of banana leaves.

Each stack contains K leaves.

Each leaf has a positive beauty value, describing how attractive it looks.

Irfan would like to take exactly P leaves to use for lunch today. If he would like to take a leaf in a stack, he must also take all of the leaves above it in that stack as well.

Help Irfan pick the P leaves that would maximize the total sum of attractive values.

Constraints:

$1 \leq T \leq 100.$

$1 \leq K \leq 30.$

$1 \leq P \leq N * K.$

$1 \leq N \leq 50.$

Input Format:

The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the three integers N, K and

P. Then, N lines follow. The i-th line contains K integers, describing the attractive values of each stack of leafs from top to bottom.

Output Format:

Print the output in a separate line contains the maximum total sum of attractive values that Irfan could pick.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
2 4 5
10 10 100 30
80 50 10 50
3 2 3
80 80
20 20
15 50
```

EXPECTED OUTPUT

```
250
180
```

Test Case 2

INPUT (STDIN)

```
2
2 4 5
20 20 70 30
60 30 10 50
3 2 3
60 70
15 40
20 10
```

EXPECTED OUTPUT

```
200
150
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int max(int a,int b)
```

Test Case 2

KEYWORD

```
for(int i = 0;i < n;i++)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

380

Test Case 3

NLOC

47

Sorting9.c +

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int max(int a, int b) {
5     return (a > b) ? a : b;
6 }
7
8 int main() {
9     int t;
10    if (scanf("%d", &t) != 1) return 0;
11    while (t--) {
12        int n, k, p;
13        scanf("%d %d %d", &n, &k, &p);
14        int dp[1505];
15        memset(dp, 0, sizeof(dp));
16        for (int i = 0; i < n; i++) {
17            int stack_vals[35];
18
19            pref[0] = 0;
20            for (int j = 1; j <= k; j++) {
21                scanf("%d", &stack_vals[j]);
22                pref[j] = pref[j - 1] + stack_vals[j];
23            }
24            for (int cur_p = p; cur_p >= 0; cur_p--) {
25                for (int x = 1; x <= k && x <= cur_p; x++) {
```

main | CH.SC.U4CSE25225 | Sorting9.c

Ln 34, Col 1 Spaces: 4 UTF-8 CRLF C

```
26         dp[cur_p] = max(dp[cur_p], dp[cur_p - x] + pref[x]);
27     }
28 }
29 }
30 printf("%d\n", dp[p]);
```

```
31  }  
32  return 0;  
33}  
34
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2
```

```
2 4 5
```

```
10 10 100 30
```

```
80 50 10 50
```

```
3 2 3
```

```
80 80
```

```
20 20
```

```
15 50
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting9.exe
```

```
250
```

```
180
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
2 4 5
```

```
20 20 70 30
```

```
60 30 10 50
```

```
3 2 3
```

```
60 70
```

```
15 40
```

```
20 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting9.exe
```

```
200
```

```
150
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Sorting	Question Information	Level 1	Challenge 20
----	---------	---------	----------------------	---------	--------------

Problem Description:

Real estate is property consisting of land and the buildings on it, along with its **natural resources** such as crops, minerals or water; immovable property of this nature; an interest vested in this (also) an item of real property, (more generally) buildings or **housing** in general.

There are 'n' flats in the village Nelvayal. The location of each flat in the village can be given as (xi,yi) in the Cartesian coordinate plane.

There are "hi" persons living in the i-th flat. Central electricity authority of the village is set to built a wire line across the village.

The wire line is supposed to constructed in a way such that it is the north-east direction.

In other words the wire line is parallel to the line y=x.

Given that the construction of such line is considered to be effective only if the number of persons living in its left and right side are equal, can you tell if the construction of such wire line is possible?

Constraints:

1 <= t <= 100
2 <= n <= 2*10^3
-10^3 <= xi, yi <= 10^3
1 <= hi <= 10^3

It is guaranteed that no two flats are at the same location.

Input Format:

The first line contains a single integer "t" denoting the number of test cases.

The first line of each test case contains 'n' i.e the number flats in the village.

Next 'n' lines contains 3 space-separated integers xi, yi, hi

Output Format:

Print the output in a separate lines contains 't' lines each containing a "YES" or "NO"

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
3	3
3	3

```
-2 1 1
-1 1 3
1 -1 4
3
-2 1 1
-1 1 1
-2 1 1
4
-2 1 1
-1 1 2
-2 1 1
3 1 5
```

EXPECTED OUTPUT

```
YES
NO
NO
```

```
1 -1 4
-1 1 3
1 -1 4
3
1 -1 4
-1 1 1
1 -1 4
4
1 -1 4
-1 1 2
1 -1 4
3 1 5
```

EXPECTED OUTPUT

```
NO
NO
NO
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t-->0)
```

Test Case 2

KEYWORD

```
for(int i = 0; i < n; i++)
```

Test Case 3

KEYWORD

```
while(l<= r)
```

Test Case 4

KEYWORD

```
int mid = (l+r)/2;
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

14

Test Case 2

TOKEN COUNT

365

Test Case 3

NLOC

60

```
Sorting10.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Flat {
5     int x;
6     int y;
7     int h;
8     int d;
9 };
10
11 int cmp_flat(const void *a, const void *b) {
12     struct Flat *f1 = (struct Flat*)a;
13     struct Flat *f2 = (struct Flat*)b;
14     return (f1->d - f2->d);
15 }
16
17 int main() {
18     int t;
19     if (scanf("%d", &t) != 1) return 0;
20     while (t-- > 0) {
21         int n;
22         scanf("%d", &n);
23         struct Flat flats[2005];
24         int total_h = 0;
25         for (int i = 0; i < n; i++) {
```

```

26         scanf("%d %d %d", &flats[i].x, &flats[i].y, &flats[i].h);
27         flats[i].d = flats[i].y - flats[i].x;
28         total_h += flats[i].h;
29     }
30     if (total_h % 2 != 0) {
31         printf("NO\n");
32         continue;
33     }
34     qsort(flats, n, sizeof(struct Flat), cmp_flat);
35     int target = total_h / 2;
36     int l = 0, r = n - 1;
37     int found = 0;

```

main | CH.SC.U4CSE25225 | Sorting10.c

Ln 61, Col 1 Spaces: 4 UTF-8 CRLF C

```
38     int cur_sum = 0;
39     for (int i = 0; i < n; i++) {
40         cur_sum += flats[i].h;
41         if (i < n - 1 && flats[i].d != flats[i + 1].d) {
42             if (cur_sum == target) {
43                 found = 1;
44                 break;
45             }
46         }
47     }
48     while (l <= r) {
49         int mid = (l + r) / 2;
```

```
50         if (mid == 0) break;
51         l = r + 1;
52     }
53     if (found) {
54         printf("YES\n");
55     } else {
56         printf("NO\n");
57     }
58 }
59 return 0;
60}
61
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

3

-2 1 1

-1 1 3

1 -1 4

3

-2 1 1

-1 1 1

-2 1 1

4

-2 1 1

-1 1 2

-2 1 1

3 1 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting10.exe

YES

NO

NO

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3

3

1 -1 4

-1 1 3

1 -1 4

3

1 -1 4

-1 1 1

1 -1 4

4

1 -1 4

-1 1 2

1 -1 4

3 1 5

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting10.exe
```

```
NO
```

```
NO
```

```
NO
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>|
```